



Programming Fundamentals

Course Work 06

TAX Calculator

Case Study

Taxation is done in different ways in several categories. It is difficult to calculate manually and you are assigned to create a tax calculator to make it easier. When the amount earned under each tax category is input, the system should calculate the amount of tax to be paid and give it as output. Also, this system has the ability to calculate leasing payments.



Figure 1 – Use case Diagram

1. Withholding Tax

There are three categories of Withholding Tax. When the user selects the Withholding Tax category correctly, the user can see the following output related to the category. The withholding Tax options are given below (Figure 3) and the user can enter an option number that he wants to execute.

```
+-----+  
|               WITHHOLDING TAX               |  
+-----+  
  
[1] Rent Tax  
  
[2] Bank Interest Tax  
  
[3] Dividend Tax  
  
[4] Exit  
  
Enter an option to continue ->
```

Figure 3 - Withholding TAX options


```
+-----+
|                                |
|                                |
+-----+
|                                |
|                                |
+-----+

Enter your rent      : -200000

      Invalid input...

Do you want to enter the correct value again (Y/N): _
```

Figure 5 – User Input Validate

```
+-----+
|                                |
|                                |
+-----+

Enter your rent      : 50000

      You dont have to pay rent tax...

Do you want to calculate another Rent Tax (Y/N) :
```

Figure 6 – When the service charge is less than Rs.100 000

II. Bank Interest Tax ([Demo](#))

Bank Interest Tax is charged from the annual bank interest. Bank Interest Tax is 5% of annual interest. When the user inputs the annual bank interest amount, the system should count the amount of the Bank Interest Tax and display it (Figure 7). Values should be validated as previously. Once calculate the Tax value is correct, it should prompt whether to calculate Bank Interest Tax again or go back to the main menu.

```
+-----+
|                BANK INTEREST TAX                |
+-----+
Enter your bank interest per year      : 50000
You have to pay Bank Interest Tax per year : 2500.00

Do you want to calculate another Bank Interest Tax (Y/N) : _
```

Figure 7 – Bank Interest Tax

III. Dividend Tax ([Demo](#))

Dividend Tax is a tax that is collected from stock market dividends. In Dividend Tax, 14% is charged for all dividends above Rs.100 000, and no tax is charged up to Rs.100 000. If the dividend value is less than Rs.100 000, print “You don’t have to pay Dividend Tax....”. Values should be validated as previously. Once calculate the Tax value is correct, it should prompt whether to calculate Dividend Tax again or go back to the main menu.

```
+-----+
|                DIVIDEND TAX                |
+-----+
Enter your total dividend per year      : 150000
You have to pay Dividend Tax per year : 7000.00

Do you want to calculate another Dividend Tax (Y/N):
```

Figure 8 – Dividend Tax

DIVIDEND TAX

Enter your total dividend per year : 50000

You dont have to pay Dividend Tax...

Figure 9 – When dividend TAX less than Rs.100 000

2. Payable Tax ([Demo](#))

The company keeps a percentage of an employee's salary and pays it as a tax, it is called Payable Tax. The percentage of monthly Payable Tax is given below.

- Payable Tax is charged above Rs.100 000 and no tax is charged up to Rs.100 000.
- Rs.100 000 to the 1st Rs.41 667 (100 000 – 141 667) – 6%
- Rs.141 667 to the 2nd Rs.41 667 (141 667 – 183 333) – 12%
- Rs.183 333 to the 3rd Rs.41 667 (183 333 – 225 000) – 18%
- Rs. 225 000 to the 4th Rs.41 667 (225 000 – 266 667) – 24%
- Rs.266 667 to the 5th Rs.41 667 (266 667 – 308 333) – 30%
- More than Rs.308 333 – 36%

Monthly Income	Tax rate	Tax
100,000	NIL	
141,667	6%	2,500
183,333	12%	7,500
225,000	18%	15,000
266,667	24%	25,000
308,333	30%	37,500
350,000	36%	52,500
450,000	36%	88,500
550,000	36%	124,500
650,000	36%	160,500
750,000	36%	196,500
900,000	36%	250,500
1,000,000	36%	286,500

Several monthly incomes and the total Payable Tax are given in the above table.

Ex:- If an employee receives Rs.900 000 salary per month, his total amount of monthly Payable Tax is Rs.250 500.

Values should be validated as previously. Once calculate the Tax value is correct, it should prompt whether to calculate Payable Tax again or go back to the main menu.

```
+-----+
|               PAYABLE TAX               |
+-----+

Enter your employee payment per month : 300000

You have to pay Payable Tax per month : 35000.00

Do you want to calculate another Payable Tax (Y/N): _
```

Figure 10 – Payable Tax

```
+-----+
|               PAYABLE TAX               |
+-----+

Enter your employee payment per month : 50000

You don't have to pay Payable Tax...

Do you want to calculate another Payable Tax (Y/N):
```

Figure 11 – When Payable TAX less than Rs.100 000

4. Social Security Contribution Levy (SSCL) Tax ([Demo](#))

A Social Security Contribution Levy (SSCL) is a payable tax imposed on importers, manufacturers, service providers, wholesalers, and retailers. SSCL Tax calculates in two steps. The steps are given below.

Step 1

Add sale tax to the value. (Sale Tax -2.5%)

Ex:- value of Goods -100 000

Sale tax - $100\,000 \times 2.5\% = 2\,500$

Value of Goods (after sale tax added) - $100\,000 + 2\,500 = 102\,500$

Step 2

Add VAT to the Value of Goods after the sale tax added.(VAT -15%)

Ex:- Value of Goods (after sale tax added) - 102 500

Value of Goods (after VAT added) - $102\,500 \times 15\% = 15\,375$

Total SSCL Tax = Sale Tax + VAT

= 2 500 + 15 375 = 17 875

User input values should be validated as previously. Once calculate the Tax value is correct, it should prompt whether to calculate SSCL Tax again or go back to the main menu.

```
+-----+
|          SOCIAL SECURITY CONTRIBUTION LEVY (SSCL) TAX          |
+-----+

Enter value of Good or Service : 100000

You have to pay SSCL Tax      : 17875.00

Do you want to calculate another SSCL Tax (Y/N):
```

Figure 13 – SSCL Tax

5. Leasing Payment

When the user input the lease amount, annual interest and the number of year, we can count monthly leasing installment by using leasing formula.

Lease Payment Calculation Formula

$$\text{Monthly installment} = \frac{A \times i}{1 - (1 / (1 + i)^n)}$$

A – lease amount

n – number of month

i – annual interest rate / 12

There are three categories of calculate Leasing Payments. When the user selects the Leasing Payment category correctly, display following output related to the category. The Leasing Payment page is given below(Figure 14) and the user can enter an option number that he wants to execute.

```
+-----+
|               Leasing Payment               |
+-----+

[1] Calculate Monthly Installment
[2] Search Leasing Category
[3] Find Leasing Amount
[4] Exit

Enter an option to continue ->
```

Figure 15 – Leasing Payment Home Page

I. Calculate Monthly Installment ([Demo](#))

First, the user should enter a valid lease amount. Otherwise, it should handle like previously. Once the user has entered a valid lease amount, the user should enter the annual interest rate and the number of years. The annual interest rate must be greater than 0% and the maximum number of the year is 5. If the year is greater than 5, then it should be notified as well as follows(Figure 17).

```
+-----+
|               Calculate Leasing Payment               |
+-----+
Enter lease amount      : 2500000
Enter annual interest rate : 22
Enter number of year    : 3
Your monthly instalment  : 95476.13

Do you want to calculate another monthly instalment (Y/N):
```

Figure 16 - Calculate Monthly Installment

Once calculate the Monthly Installment is correctly, it should prompt whether to calculate the Monthly Installment again or go back to the main menu.

```
+-----+
|               Calculate Leasing Payment               |
+-----+
Enter lease amount      : 1000000
Enter annual interest rate : 22
Enter number of year    : 6
      Invalid number of year... Enter the correct value again...

Enter number of year    : 3
Your monthly instalment  : 38190.45

Do you want to calculate another monthly instalment (Y/N):
```

Figure 17 - When input incorrect year

II. Search Leasing Category ([Demo](#))

First, the user should enter the lease amount and the annual interest rate. The annual interest rate must be greater than 0%. The system should display monthly payments of 3 years, 4 years, and 5 years of leasing plans.

```
+-----+
|                                     |
|               Search Leasing Category               |
|-----+-----+
|
| Enter lease amount           : 2500000
|
| Enter annual interest rate : 22
|
| Your monthly instalment for 3 year leasing plan - 95476.13
| Your monthly instalment for 4 year leasing plan - 78765.19
| Your monthly instalment for 5 year leasing plan - 69047.28
|
| Do you want to Search another Leasing Category (Y/N):
```

Figure 18 – Search Leasing Category

III. Find the Leasing Amount ([Demo](#))

First, the user should enter the monthly lease payment amount user can pay, the annual interest rate, and the number of years. The annual interest rate must be greater than 0% and the maximum number of the year is 5. Finally, the system should calculate the total leasing amount available from the leasing company.

```
+-----+
|                                     |
|               Find Leasing Amount               |
|-----+-----+
|
| Enter the monthly lease payment amount you can afford : 50000
|
| Enter number of year           : 3
|
| Enter annual interest rate     : 22
|
| You can get Lease Amount       : 1309228.00
|
| Do you want to calculate another monthly instalment (Y/N): _
```

Figure 19 – Find Leasing Amount

Guideline

- Refer to the Coursework Guidelines at the end to understand the specific guidelines to be followed when developing the project required.
- You can't create classes except for the class that holds the main method.
- Use the Scanner class to get input from the command-line-interface.
- All validations that have been mentioned in this document should be implemented.
- The code to clear the command line from inside a Java application is as follows. You can use this code when you need to clear the command line.

```
public final static void clearConsole() {  
    try {  
        final String os = System.getProperty("os.name");  
        if (os.contains("Windows")) {  
            new ProcessBuilder("cmd", "/c", "cls").inheritIO().start().waitFor();  
        } else {  
            System.out.print("\033[H\033[2J");  
            System.out.flush();  
        }  
    } catch (final Exception e) {  
        e.printStackTrace();  
        // Handle any exceptions.  
    }  
}
```

- You can create as many methods as you wish in the only class that you have.
- Demo Videos are given at relevant places for you to understand the coursework requirement better and Demo videos may help you to clarify your doubts to some extent.
- Submission – You should attach your Java file with the file name format [BatchNumber_ Name], and submit on or before the deadline.
- If you still have any questions, feel free to question.